

District Navigation Mobile Application

Project Overview

The application is a mobile navigation solution designed specifically for residential districts and housing projects. Users can search for building numbers within the district and instantly launch Google Maps navigation to their destination.

The geographic data is managed through Google Earth Pro using KML files, allowing administrators to add or update buildings without requiring users to download a new version of the application.

The proposed architecture provides a lightweight, scalable, and cost-effective solution with no backend server requirements and full offline capabilities.

Core Features

Building Search

Users can:

- Search for buildings by number.
 - View building information.
 - Access frequently used or popular buildings.
 - Instantly navigate to their selected destination.
-

Google Maps Navigation

The application does not implement its own navigation engine.

Instead, it integrates with Google Maps and launches turn-by-turn navigation using the building coordinates stored in the district data.

This provides:

- Real-time traffic information
 - Optimized routes
 - Voice guidance
 - Familiar user experience
-

Automatic Data Synchronization

The system uses a hash-based synchronization mechanism to keep the application data up to date without requiring Play Store updates.

Administrator Workflow

Google Earth Pro



Update District Buildings



Export Updated KML File



Generate SHA-256 Hash



Upload:

- district.kml
- district.hash

to Google Drive

Mobile Application Workflow

Application Startup



Download Remote Hash



Compare With Local Cached Hash



If Different:

- Download New KML
- Replace Local Cache
- Parse KML
- Generate Local JSON Cache



If Identical:

- Use Existing Local Data

This approach minimizes bandwidth usage while ensuring users always have the latest district information.

Offline Support

The application stores synchronized district data locally on the device.

Users can:

- Search for buildings
- Access previously downloaded data
- Launch navigation

even when internet connectivity is unavailable.

Internet access is only required when checking for new district updates.

Advertisement Monetization

The application will integrate Google AdMob to generate revenue.

Planned monetization features include:

Banner Advertisements

Non-intrusive advertisements displayed within the application interface.

Optional Interstitial Advertisements

Full-screen advertisements displayed periodically during user interactions while maintaining a positive user experience.

Technical Architecture

Geographic Data Management

Technology:

- Google Earth Pro
- KML (Keyhole Markup Language)

The district administrator manages all building locations through Google Earth and exports updated KML files whenever changes are made.

Cloud Storage

Technology:

- Google Drive

Stored Files:

- district.kml
- district.hash

Google Drive acts as a lightweight distribution mechanism, eliminating the need for custom backend infrastructure.

Mobile Application

Technology:

- Flutter
- Material Design 3

Platform Support:

- Android (Initial Release)
 - iOS (Future Expansion)
-

Local Storage

The application maintains:

- Cached KML files
- SHA-256 hash values
- Generated JSON search data

This provides:

- Fast startup times
 - Offline functionality
 - Reduced network traffic
-

Navigation Provider

Technology:

- Google Maps

The application launches Google Maps directly using geographic coordinates, leveraging Google's navigation infrastructure instead of implementing custom routing solutions.

User Interface Design

The application follows a search-first design philosophy optimized for quick building lookup and navigation.

Example Flow:

Search Building Number

↓

Display Building Information

↓

View On Map

↓

Start Navigation

This simple workflow minimizes user effort while providing immediate access to navigation services.

Scalability

The proposed architecture supports:

- Additional buildings
- New residential projects
- Larger district datasets
- Future expansion to multiple districts

without requiring application updates or backend modifications.

Future Enhancements

Potential future features include:

- Nearby services and facilities
- Schools
- Hospitals
- Mosques
- Real-estate advertisements
- Favorite locations
- Recently visited buildings
- Analytics and usage statistics
- Multi-district support
- iOS deployment

The current architecture allows these enhancements to be implemented incrementally while maintaining low operational costs.

Benefits

The proposed solution provides:

- No backend server costs
- No database maintenance
- No infrastructure management
- Automatic updates
- Offline capabilities
- Google Maps integration
- Advertisement monetization
- Long-term scalability
- Easy management through Google Earth Pro